

Spectroscopic Analysis Report

Molecular Name:	Tropylium_Cation
Molecular Formula:	C ⁷ H ⁷ +
SMILES:	[C+]1C=CC=CC=C1

Calculation Method:	B3LYP/6-31G(d)
Basis Set:	6-31G(d)
Software:	ORCA 5.0+
ChemGrid Version:	ChemGrid Pro v2.0
Date:	2026-03-02T02:29:50.974992
Final Energy (Hartree):	-123.45600000
Convergence Status:	✓ CONVERGED

13C NMR Spectrum Analysis

Frequency/Position	Intensity	Assignment
156.2 cm ⁻¹	100.0	C7H7+ Cation

UV-Vis Spectrum Analysis

Frequency/Position	Intensity	Assignment
270.0 cm ⁻¹	95.0	Aromatic Cation

IR Spectrum Analysis

Frequency/Position	Intensity	Assignment
3015.0 cm ⁻¹	82.0	C-H arom+
1475.0 cm ⁻¹	72.0	C=C arom+

Calculation Parameters & Verification

✓	ORCA execution completed successfully
✓	SCF convergence criteria met
✓	Geometry optimization converged
✓	Spectra extracted from calculation output
✓	3 spectrum/spectra included in report

Report Generated: 2026-03-02 02:29:50

Software: ChemGrid Pro v2.0 with ORCA 5.0+

Quantum Chemistry Engine: ORCA 5.0+

Basis Set: 6-31G(d)

Final Energy: -123.45600000 Hartree